

Ricardo (Zhicong) Xie

Mechanical Engineering Intern | CAD/BIM, Manufacturing, HVAC and Project Support | Open NZ-wide

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LinkedIn: <https://www.linkedin.com/in/ricardo-xie-6394643b4> | Portfolio: <https://zxie588-alt.github.io> | GitHub: <https://github.com/zxie588-alt>

NZ student visa | Full-time eligible over scheduled summer break 17 Nov 2026 - late Feb 2027 | No employer visa support needed for this internship | Open NZ-wide

PROFILE

- Mechanical engineering master's student open to internships across New Zealand in mechanical, HVAC, manufacturing, CAD/BIM or project support.
- Practical evidence includes Revit MVHR/ERV, SolidWorks enclosure design, AutoCAD/DXF workshop work, FEA/CFD and STM32 firmware.
- Portfolio with project screenshots, source files and Revit/MVHR evidence: <https://zxie588-alt.github.io>

EDUCATION

University of Auckland - Master of Engineering Studies, Mechanical Engineering | 2026 - Present

Study focus and portfolio work in mechanical, building-services, manufacturing and analysis contexts.

Study record/focus: Completed: Computational Fluid Dynamics, Composite Materials, Engineering Methods, Aerospace Systems Engineering.

Registered S2: Building Services, Smart Manufacturing, Manufacturing and Industrial Systems, Research Project.

Huazhong Agricultural University - Bachelor of Engineering, Mechanical Engineering / Mechatronics | 2021 - 2025

Mechanical design, CAD, electrotechnics, sensors, embedded control and mechatronic system integration.

PRACTICAL AND TEAM EXPERIENCE

- Engineering team leadership: led task scoping, workload split and final technical presentation in team design work.
- Workshop practice: prepared AutoCAD/DXF laser-cut geometry and checked cut parts against drawings.
- Hands-on exposure: CNC machining, workholding, cutting fluid, chip formation and 3D printing workflow.
- Lab practice: interpreted force-displacement tests and controlled a robotic arm in training sessions.
- Technical communication: former student news lead; strong project records, photography and presentation visuals.

PROJECT WORK

Small Residential MVHR / ERV Revit MEP Coordination

Revit 2026 MEP coordination, MVHR/ERV layout, airflow schedule, SolidWorks, AutoCAD/DXF, NZBC G4/H1 context, commissioning notes

- Built a small NZ residential MVHR/ERV coordination package with Revit MEP, SolidWorks and AutoCAD/DXF evidence.
- Created plan, 3D view, airflow schedule, sheet, review comments and simple clash/coordination notes.
- Documented preliminary airflow, pressure-loss, balancing and commissioning actions for the next design iteration.

Manufacturing Workshop Training & DXF Preparation

Laser cutting, AutoCAD/DXF, CNC machining exposure, 3D printing workflow

- Prepared AutoCAD/DXF geometry for laser cutting and checked the cut plate against drawing intent.
- Practised CNC workflow: workholding, tool motion, cutting fluid, chip formation and shop safety.
- Reviewed machined gears and shafts for tolerance, edge quality and tool-access constraints.

Ruggedized Outdoor IoT Sensor Module Design Package

SolidWorks, enclosure design, DFM notes, Keil C, testing workflow

- Designed a SolidWorks enclosure with gasket stack-up, screw bosses, PCB envelope and service access.
- Prepared DFM notes for CNC review, 3D printing and assembly tolerance checks.
- Reviewed load paths around corner, strap-lug and screw-boss regions; built Keil C module structure.

Composite Formula SAE Front Wing Analysis

Abaqus/Standard, shell FEA, composite layup, structural checks

- Built an Abaqus shell FEA model with S4R elements, laminate layups and local mesh refinement.
- Converted aero, lateral cornering and 4.1 G bump conditions into structural load cases.
- Checked ply-level stress use; peak utilisation was about 74% under the 4.1 G bump case.

TECHNICAL SKILLS

- CAD/BIM: Revit 2026 MEP, SolidWorks assemblies, AutoCAD/DXF, STEP/STL export.
- Hands-on: laser cutting, CNC workflow, 3D printing workflow, drawing-to-part checking.
- Building services: MVHR/ERV layout, airflow schedule, duct velocity and pressure checks.
- Analysis: Abaqus FEA, ANSYS Fluent/CFX, thermal-fluid modelling and verification notes.
- Coordination: team leadership, review comments, risk register and technical documentation.